

An aerial photograph of a dense urban area in Singapore, showing numerous high-rise public housing blocks (HDB flats) with white and orange facades. The buildings are arranged in a grid-like pattern, with some greenery and trees interspersed between them. The sky is a mix of blue and orange, suggesting a sunset or sunrise. The overall scene is a typical representation of Singapore's public housing landscape.

HDB RESALE MARKET ANALYSIS & PREDICTIVE PRICING MODEL IN SINGAPORE

Presented By:

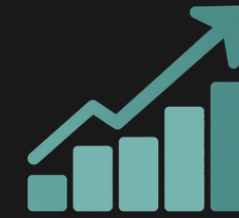
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EXECUTIVE SUMMARY



PROJECT GOALS

Analyze HDB resale market trends, identify strongest drivers of resale prices and build a predictive pricing model for data-driven decisions.



MARKET INSIGHTS

Identified key pricing factors including location, flat type, floor level, lease age and MRT proximity.



MODEL ACCURACY

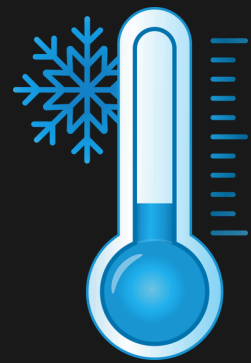
Achieved high predictive accuracy with R^2 of 0.90, enabling reliable price estimations.



BUSINESS VALUE

Empowers agency with data-driven pricing to better serve clients and optimize transactions in a competitive market

MARKET OVERVIEW



MARKET CORRECTION & COOLING MEASURES 2013–2017

Government cooling measures **significantly moderated speculative demand** and contributed to a multi-year correction in HDB resale prices after the rapid growth observed in the early 2010s.



RECOVERY & STABILIZATION OF THE HDB RESALE MARKET 2018–2019

By 2018–2019, the HDB resale market showed signs of stabilization, with **accessibility, location, and flat size** emerging as increasingly important pricing differentiators.



PANDEMIC-ERA SURGE & RESALE DEMAND BOOM 2020–2021

The pandemic significantly reshaped buyer priorities, **accelerating demand for larger and immediately available resale flats**, which contributed to a strong upward price trend during 2020–2021.



Singapore's HDB resale market evolved from a cooling-measure-driven correction phase into a stabilized recovery period before experiencing a strong pandemic-era demand surge, highlighting how **government policy, economic conditions, and changing buyer behavior collectively shaped resale price trends**.

WOW! MANAGEMENT



Adrian Koh

Director of Strategy & Growth at WOW!

PROFILE

- Oversees agency competitiveness and digital transformation
- Wants data-driven differentiation in the property market

OUR BUSINESS VALUE:

- Demonstrates business value of predictive analytics
- Enhances WOW!'s market positioning & competitiveness



Rachel Tan

Senior Property Consultant at WOW!

PROFILE

- 35 years old
- Handles HDB resale buyers and sellers across Singapore
- Works in a highly competitive market
- Needs to provide fast and accurate pricing advice to clients

OUR BUSINESS VALUE:

- Predictive model gives reliable estimated resale prices
- Insights identify price drivers to support client presentations (MRT, lease age, floor area, planning area, flat-type)
- Faster, more confident pricing recommendations

OUR CUSTOMERS



Mr. Lim

Existing HDB Owner Looking to Sell

PROFILE

- 48 years old
- Owns a 4-room flat in Tampines
- Wants to maximize selling price
- Concerned about market timing

OUR BUSINESS VALUE:

- Explains pricing factors transparently
- Provides evidence-based pricing ranges
- Helps optimize listing strategy



Sarah & Daniel

Young Couple Buying Their First HDB

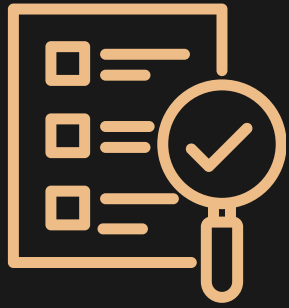
PROFILE

- Early 30s
- Budget-conscious
- Comparing multiple towns
- Concerned about long-term value

OUR BUSINESS VALUE:

- Shows relationship between price and amenities
- Helps evaluate MRT accessibility vs cost
- Supports smarter purchase decisions

DATA & METHODOLOGY



DATASET OVERVIEW

- 150,634 HDB Resale Transactions
- 77 Columns
- Transaction Dates between March 2012 to April 2021



DATA CLEANING & PRE-PROCESSING

Data Quality Verification performed:

1. Uniqueness of transaction id
2. Transactions with planning_area is not the same or not logical with town removed
(Example: Kallang/Whampoa)
3. Consistency check between HDB_age and Year of Completed Construction
4. Correctness of floor level across various fields such as storey_range, lower, upper and mid

DATA LIMITATIONS

1. Market Conditions & Macroeconomic Factors Not Included

External factors such as **interest rates, inflation, cooling measures, economic sentiment** may influence resale prices significantly.

2. Property Condition & Renovation Quality Not Captured

Qualitative factors such as **renovation quality, interior condition, facing/view, unit layout efficiency, and maintenance condition** may affect actual transaction prices.

3. Correlation Does Not Equal Causation

While features such as MRT proximity and mall accessibility show strong relationships with price, they may also be influenced by **broader location desirability and urban development factors**.

4. Geographic Bias in Transaction Volume

Certain planning areas have **significantly more transactions than others**, which may influence the strength and reliability of insights across regions.

Example: Tampines, Jurong West, Sengkang vs Tanglin and Downtown Core

4. Model Predictions Are Estimates

The predictive model should be used as a **pricing support tool**. Human expertise and market judgment remain important.

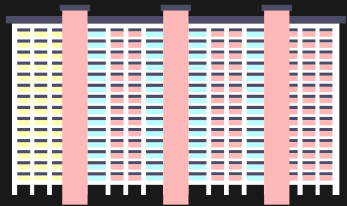
 Despite these limitations, the analysis provides **strong directional insights** into the key drivers of HDB resale prices and offers a robust foundation for **data-driven pricing decisions**.

EXPLORATORY ANALYSIS



PLANNING AREA

Central and mature planning areas such as Outram and Downtown Core command the highest HDB resale prices due to **strong accessibility, connectivity, and urban amenities**.



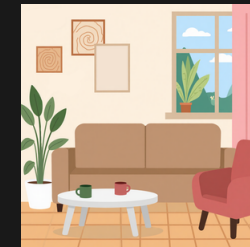
FLAT TYPE

Executive and Multi-Generation flats consistently achieved the highest resale prices, reflecting sustained demand for larger housing units.



FLOOR LEVEL

Higher-floor HDB units command stronger resale premiums as buyers value better views, privacy, and ventilation.



LEASE AGE

Younger HDB flats generally achieve higher resale prices due to longer remaining leases and stronger buyer preference.



Team's Hypothesis: Singapore's HDB resale market is primarily driven by **location, flat type, floor level, and housing age**, with premium segments consistently outperforming the broader market.

FEATURE IMPORTANCE

TOP PREDICTORS

Features that were identified for the model:

1. Planning Area/Neighbourhood
2. Floor level
3. Flat Type
4. HDB Age
5. Distance to nearest MRT
6. Distance to nearest Bus Stop
7. Distance to nearest Primary School

WHY THESE MATTER?

These are the common metrics that potential home buyers are typically concerned with.

The principles can be summarized based on:

- Size
- Convenience
- Future Planning
- Accessibility

PREDICTIVE MODELING

MODELS TESTED

- Linear Model
- Decision Tree

METRICS COMPARISON

- Root Mean Squared Error
- R Squared Value

	Linear Model	Decision Tree Depth: 30	Interpretation
Root Mean Square Error	\$61k	\$43K	The range of values of the prediction from the true value
R Squared Value	81/100	90/100	100 is the model predicts the price perfectly. 0 would be akin to taking the average price of all houses as your prediction

💎 The **Decision Tree Model** outperformed the Linear Model with a **lower RMSE (~\$43K)** and **higher R² score (90%)**. An **optimal tree depth of 30** best captured pricing patterns while balancing model accuracy and overfitting risk.

Note: The Decision Tree Model would still require further testing and fine-tuning for unlikely cases.

RECOMMENDATIONS



BUILD A DATA-DRIVEN PRICING ADVISORY FRAMEWORK

- Leverage **predictive analytics** to provide agents with faster, more accurate, and data-backed pricing recommendations for HDB resale flats.
- Improve **pricing consistency, strengthen client trust, and enhance competitiveness** in the resale market.
- Utilize **Linear Regression** as an initial analysis tool.



LEVERAGE ANALYTICS AS A COMPETITIVE ADVANTAGE

- Position as a **technology-enabled real estate agency** by integrating predictive pricing tools and market intelligence into its services.
- Creates opportunities for **future AI-driven valuation platforms, smarter client engagement, and stronger differentiation in the property market.**



PRIORITIZE LOCATION & CONNECTIVITY INSIGHTS

- **Location and MRT accessibility** are among the strongest drivers of resale prices, making them critical factors in client advisory and negotiations.
- Empower agents with **planning area and connectivity insights to better justify pricing premiums and market positioning.**

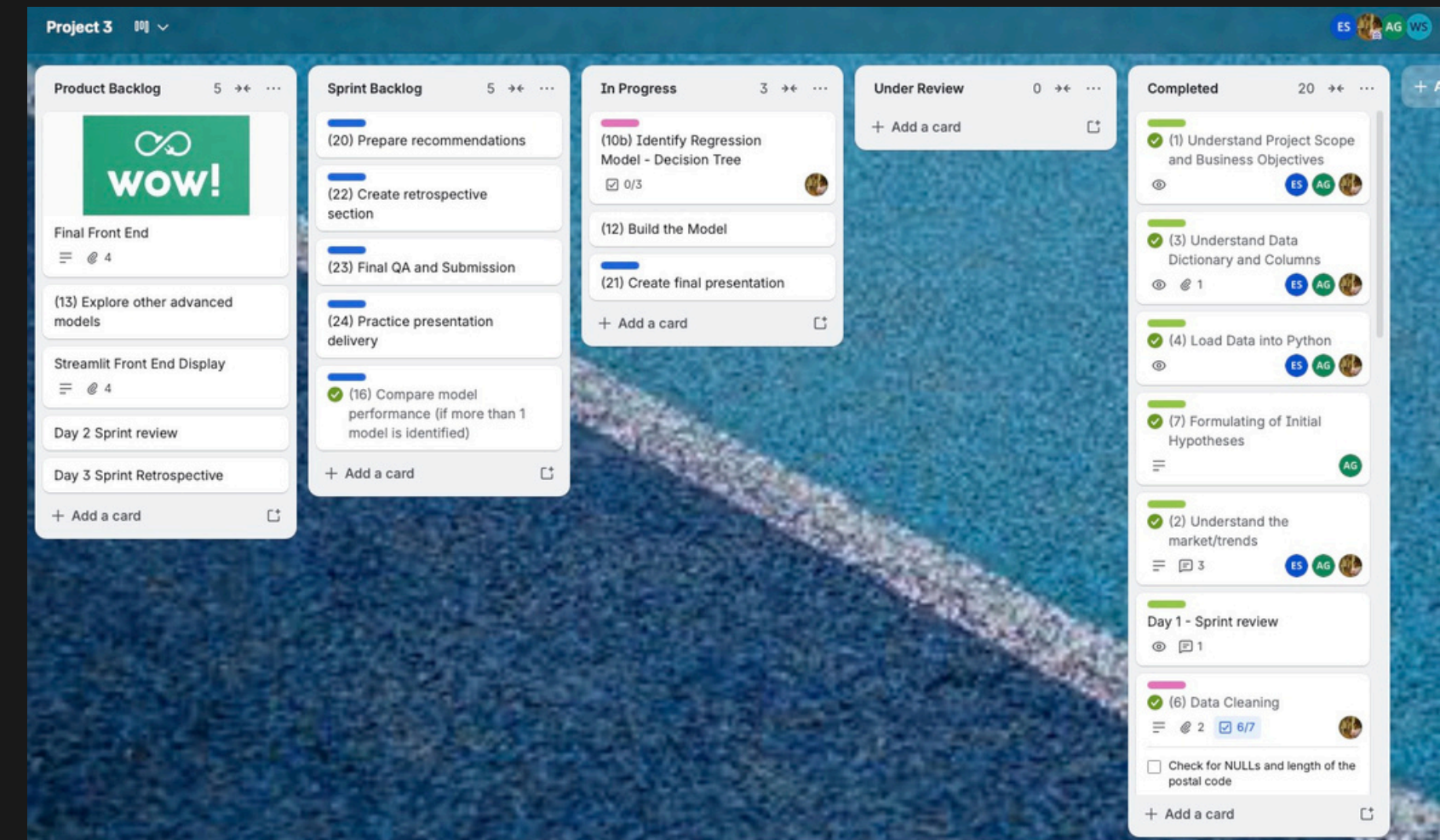
AGILE SPRINT RETROSPECTIVE



KEY LEARNINGS

- **Coordinating with teammates** to understand work flow progression and how to combine completed / in-progress works
- Benefits of having **thought partners** to approach the same work differently
- **Graphic User Interface and Backend Integration** has unique challenges when the predictive model requires one-hot encoded information.
- Importance of Team Coordination to use the **same data set** post cleaning for visualizations and modelling.
- Better **understand of the deviation between the Linear Regression Model and the Decision Tree Model**. Although the Decision Tree R2 Value is good, the RSME value is still relatively high and the Decision Tree model estimates certain 1 room flats more expensive than a 3 room flat all things being equal.

THE TEAM'S TRELLO BOARD



DEMO OF PREDICTIVE MODEL

Agents will be able to use this tool on the go when they meet clients to perform price estimations based on the selected parameters.

Tell Me Your Dream Home

Predictive Model Preference

Linear Regression

Predicted Price:

\$571245.86

Enter Flat Type

4 Room

Enter Planning Area

Tampines

Enter Desired Floor

16

Enter Lease Age

20

Enter Distance to Nearest MRT in Meters

500

Enter Distance to Nearest Bus Stop in Meters

200

Enter Distance to Nearest Primary School in Meters

600

Confirm Preferences



THANK YOU!